PRACTICAL 6
File Ownership, Permissions and ACL

Aim
To manage Linux ownership, standard permissions, special permissions and ACLs.

A. View permissions
ls -l
Example:
-rw-r--r-- 1 student student 15 Sep 20 10:20 file.txt
Permission format:

```
- rw- r-- r--
  |  |  |
  |  |  +--- Others
  |  +------- Group
  +---------- Owner
```

B. chmod
Create file:
touch permission.txt
Set permissions:
chmod 755 permission.txt
Check:
ls -l permission.txt

Numeric Permissions

Number	Permission
0	---
1	--x
2	-w-
3	-wx
4	r--
5	r-x
6	rw-
7	rwx

Example:
chmod 644 file.txt

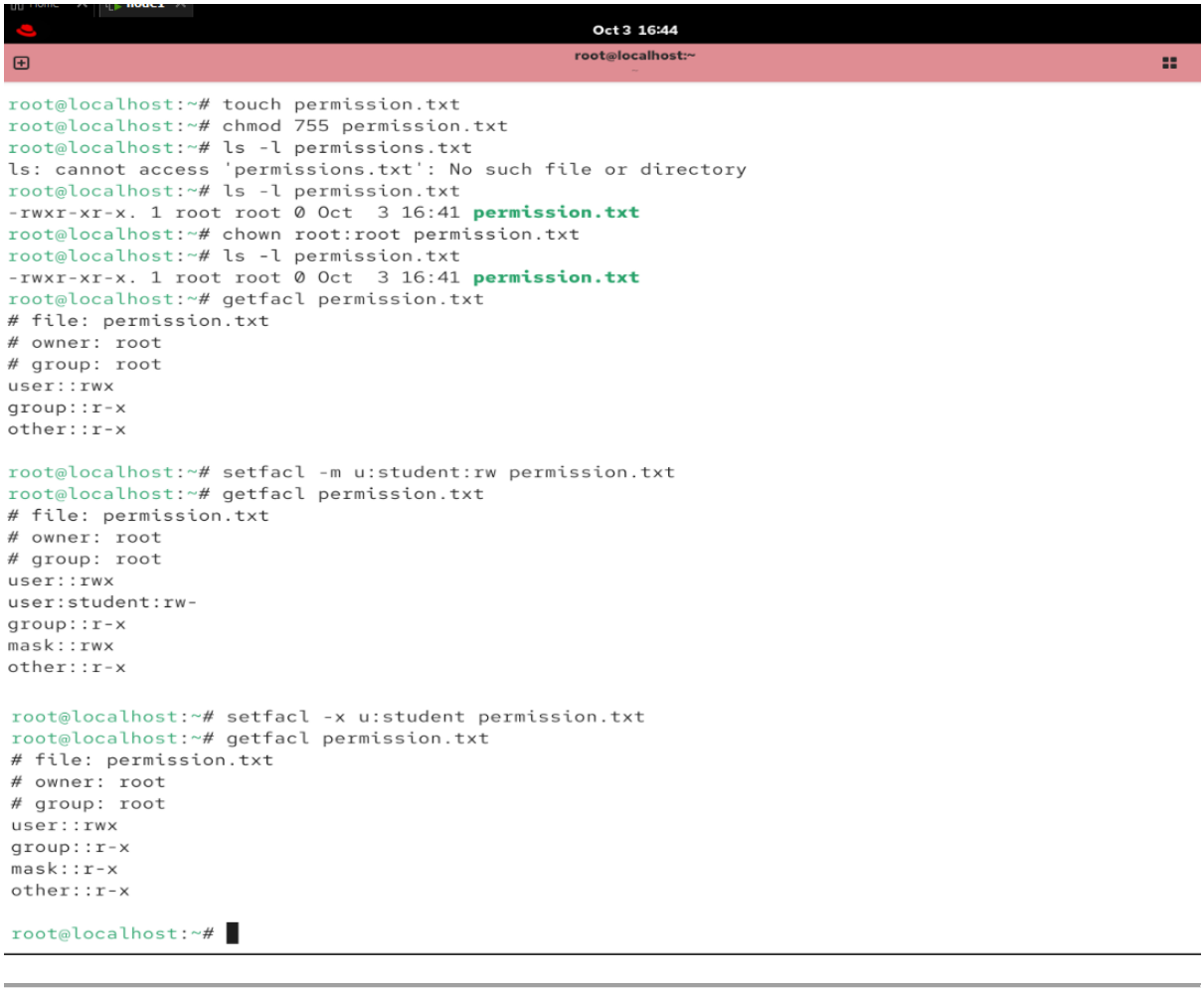
C. chown
sudo chown root:root permission.txt
Check:
ls -l permission.txt

D. umask
umask
Temporarily set:

umask 027
Create:
touch umaskfile
Check:
ls -l umaskfile

E. set-GID Directory
Create group:
sudo groupadd project
Create directory:
sudo mkdir /project
Change group:
sudo chgrp project /project
Set set-GID:
sudo chmod 2775 /project
Verify:
ls -ld /project
The group directory should show s in the group permission position.

F. ACL
Check ACL:
getfacl permission.txt
Set ACL:
sudo setfacl -m u:student:rw permission.txt
Check:
getfacl permission.txt
Remove ACL:
sudo setfacl -x u:student permission.txt


Oct 3 16:44
root@localhost:~
root@localhost:~# touch permission.txt
root@localhost:~# chmod 755 permission.txt
root@localhost:~# ls -l permissions.txt
ls: cannot access 'permissions.txt': No such file or directory
root@localhost:~# ls -l permission.txt
-rwxr-xr-x. 1 root root 0 Oct 3 16:41 permission.txt
root@localhost:~# chown root:root permission.txt
root@localhost:~# ls -l permission.txt
-rwxr-xr-x. 1 root root 0 Oct 3 16:41 permission.txt
root@localhost:~# getfacl permission.txt
file: permission.txt
owner: root
group: root
user::rwx
group::r-x
other::r-x
root@localhost:~# setfacl -m u:student:rw permission.txt
root@localhost:~# getfacl permission.txt
file: permission.txt
owner: root
group: root
user::rwx
user:student:rw-
group::r-x
mask::rwx
other::r-x
root@localhost:~# setfacl -x u:student permission.txt
root@localhost:~# getfacl permission.txt
file: permission.txt
owner: root
group: root
user::rwx
group::r-x
mask::r-x
other::r-x
root@localhost:~#

PRACTICAL 7
Users, Groups and Superuser Administration

Aim

To create and manage users/groups, configure passwords, password aging and superuser access.

Create User

sudo useradd labuser
Set password:
sudo passwd labuser

Create Group

sudo groupadd tyitgroup
Add user to group:
sudo usermod -aG tyitgroup labuser
Verify:
id labuser

Password Aging

sudo chage -l labuser
Set maximum password age:
sudo chage -M 90 labuser
Set minimum age:
sudo chage -m 1 labuser
Set warning period:
sudo chage -W 7 labuser
Verify:
sudo chage -l labuser

---

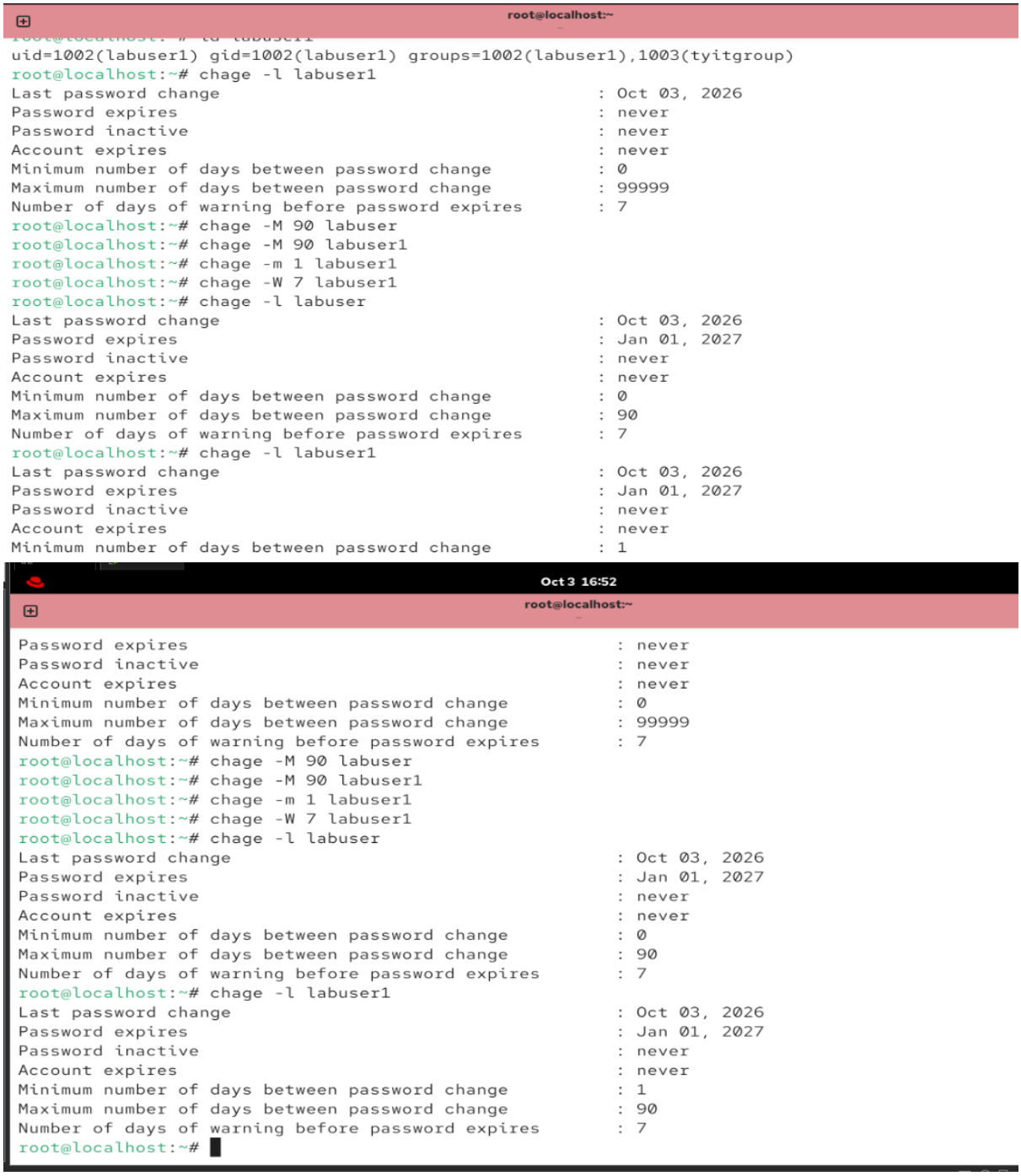
su

su - labuser
Return:
exit

---

sudo

Check:
sudo -l
Run administrative command:
sudo whoami
Expected:
root


root@localhost:~
uid=1002(labuser1) gid=1002(labuser1) groups=1002(labuser1),1003(tyitgroup)
root@localhost:~# chage -l labuser1
Last password change : Oct 03, 2026
Password expires : never
Password inactive : never
Account expires : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
root@localhost:~# chage -M 90 labuser
root@localhost:~# chage -M 90 labuser1
root@localhost:~# chage -m 1 labuser1
root@localhost:~# chage -W 7 labuser1
root@localhost:~# chage -l labuser
Last password change : Oct 03, 2026
Password expires : Jan 01, 2027
Password inactive : never
Account expires : never
Minimum number of days between password change : 0
Maximum number of days between password change : 90
Number of days of warning before password expires : 7
root@localhost:~# chage -l labuser1
Last password change : Oct 03, 2026
Password expires : Jan 01, 2027
Password inactive : never
Account expires : never
Minimum number of days between password change : 1
Oct 3 16:52
root@localhost:~
Password expires : never
Password inactive : never
Account expires : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
root@localhost:~# chage -M 90 labuser
root@localhost:~# chage -M 90 labuser1
root@localhost:~# chage -m 1 labuser1
root@localhost:~# chage -W 7 labuser1
root@localhost:~# chage -l labuser
Last password change : Oct 03, 2026
Password expires : Jan 01, 2027
Password inactive : never
Account expires : never
Minimum number of days between password change : 0
Maximum number of days between password change : 90
Number of days of warning before password expires : 7
root@localhost:~# chage -l labuser1
Last password change : Oct 03, 2026
Password expires : Jan 01, 2027
Password inactive : never
Account expires : never
Minimum number of days between password change : 1
Maximum number of days between password change : 90
Number of days of warning before password expires : 7
root@localhost:~#


root@localhost:~
Number of days of warning before password expires : 7
root@localhost:~# chage -l labuser1
Last password change : Oct 03, 2026
Password expires : Jan 01, 2027
Password inactive : never
Account expires : never
Minimum number of days between password change : 1
Maximum number of days between password change : 90
Number of days of warning before password expires : 7
root@localhost:~# su - labuser1
labuser1@localhost:~$ exit
logout
root@localhost:~# -l
bash: -l: command not found...
root@localhost:~# sudo -l
Matching Defaults entries for root on localhost:
!visiblepw, always_set_home, match_group_by_gid, always_query_group_plugin, env_reset,
env_keep="COLORS DISPLAY HOSTNAME HISTSIZE KDEDIR LS_COLORS", env_keep+="MAIL PS1 PS2 QTDIR USERNAME
LANG LC_ADDRESS LC_CTYPE", env_keep+="LC_COLLATE LC_IDENTIFICATION LC_MEASUREMENT LC_MESSAGES",
env_keep+="LC_MONETARY LC_NAME LC_NUMERIC LC_PAPER LC_TELEPHONE", env_keep+="LC_TIME LC_ALL LANGUAGE
LINGUAS _XKB_CHARSET XAUTHORITY", secure_path=/sbin\:/bin\:/usr/sbin\:/usr/bin
User root may run the following commands on localhost:
(ALL) ALL
root@localhost:~# whoami
root
root@localhost:~#